

A top-down view of a grey desk with various items: a stethoscope, a laptop displaying a medical website, a tablet, a pair of glasses, a smartphone, a spiral notebook with a pen, and a coffee cup. A large white pill-shaped graphic is centered over the laptop.

Pharm*Achieve*

CATARACT SURGERY POST-OPERATIVE CARE

Qualifying Exam Preparatory Course

COMPETENCIES COVERED IN THIS PRESENTATION...

**Providing Care:
Clinical Care**

**Providing Care:
Distribution**

**Knowledge
& Expertise**

**Communication
& Collaboration**

**Leadership &
Stewardship**

Professionalism

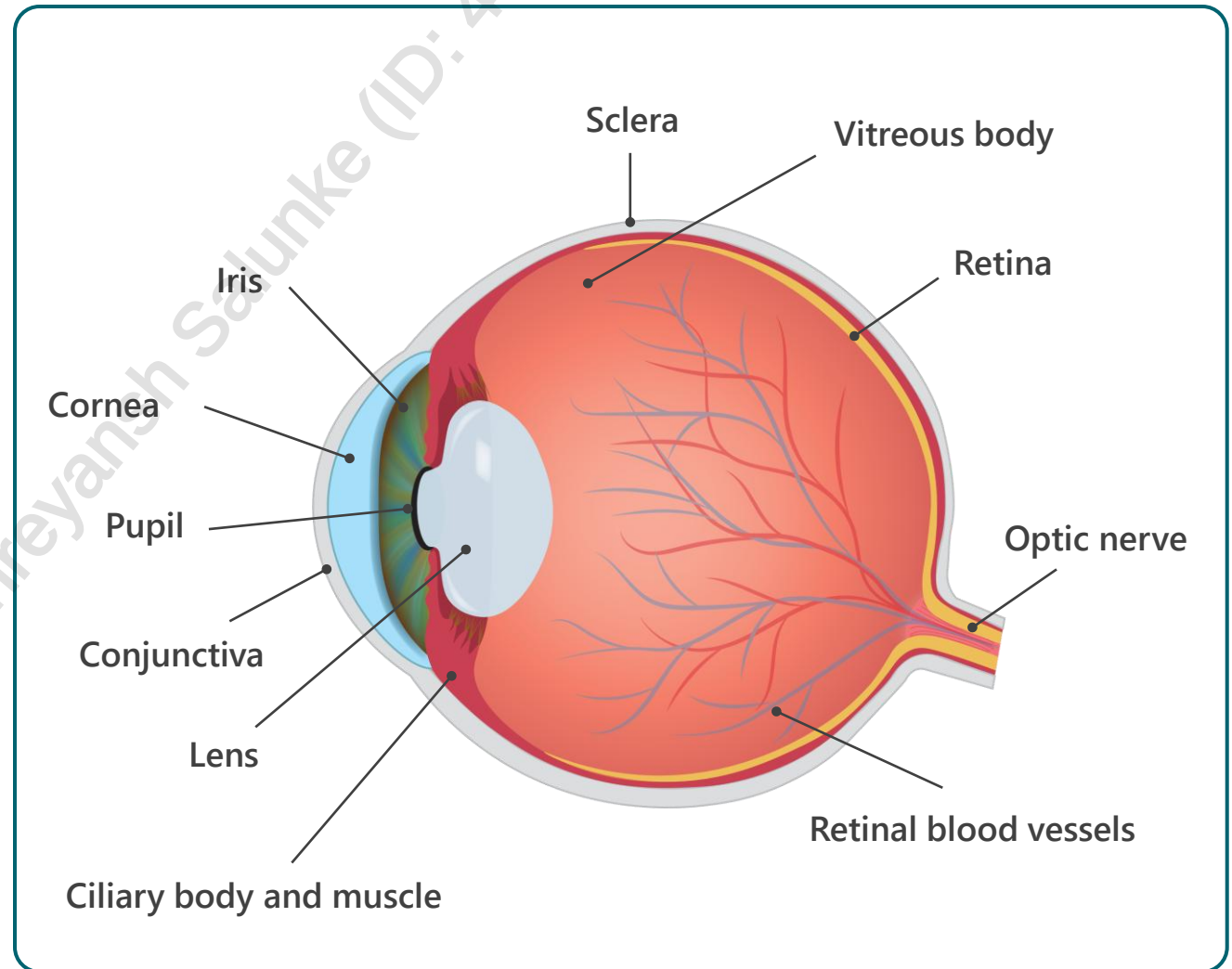
LEGEND

BID	Bis In Die (twice daily)
H	Hours
IOP	Intraocular pressure
NSAIDs	Non-steroidal Anti-inflammatory Drugs
PO	Per Os (Oral)
PRN	Pro Re Nata (As Needed)
Q	Quaque (Every)

Licensed to Shreyansh Salunke (ID: 4477)

PATHOPHYSIOLOGY

- A cataract is the **clouding of the eye lens which impacts vision**
- The lens is mostly composed of water and protein
- With increasing age, the protein may clump and start clouding the lens
- Other possible causes of lens protein clumping include smoking and diabetes
- Symptoms include:
 - **Blurred vision**, glare, seeing floaters, double vision, **trouble seeing at night**



GOALS OF THERAPY

- ✓ Perioperative prophylactic pharmacological treatment
 - ✓ Control inflammation
 - ✓ Prevent infection
 - ✓ Maintain eye comfort
 - ✓ Promote visual rehabilitation early on
- ✓ Postoperative assessment
 - ✓ Identify early-stage intraocular infections
 - ✓ Monitor for postoperative uveitis or intraocular pressure (IOP) elevation
 - ✓ Detect complications such as retinal detachment, iris prolapse, wound leak, flat anterior chamber, excessive corneal edema, or intraocular hemorrhage

POSTOPERATIVE ASSESSMENT PROCESS

Pain

- **Normal:** comfortable with minimal pain, slight foreign-body sensation
- **Abnormal:** severe pain may suggest increased IOP, inflammation, and/or infection

Trauma

- Any trauma in early phase to postoperative eye requires assessment with slit lamp

Change in vision

- Worsening, darkening, loss of detail, peripheral visual loss
 - Could indicate infection, hemorrhage, retinal detachment, or other issues requiring immediate attention

Visual phenomena

- Worsening flashing lights, dark shadows, or floaters
 - Could indicate infection, retinal tear/detachment or uveitis

POSTOPERATIVE ASSESSMENT PROCESS

Itching of the eye

- Suggests dry eye or drug allergy

Examination of the eye

- Swelling of lids/conjunctiva suggests drug allergy or infection
- Photophobia can indicate anterior uveitis or corneal haze due to infection, inflammation, or increase IOP
- Cornea should be clear of clouding or infiltrates

Review ocular medications

- Reinforce proper use
- Clarify confusion
- Discuss with caregivers

Verify follow-up appointments

PHARMACOLOGICAL OPTIONS OVERVIEW

- 1 **Topical Antibiotics:** reduce risk of endophthalmitis
- 2 **Anti-Inflammatory Agents:** decrease inflammation
- 3 **Glaucoma Agents:** lower IOP
- 4 **Dilators and Cycloplegics:** keep iris away from implant during early healing period + improve discomfort

PHARMACOLOGICAL OPTIONS

ANTIBIOTICS

- **Indication:** to reduce risk of endophthalmitis following surgery
- Can be used pre-, peri-, and post-operatively – regimens may vary by practitioner
- **Tobramycin**, besifloxacin, ciprofloxacin, gatifloxacin, moxifloxacin, ofloxacin
- **Side effects:**
 - All: blurred vision, local irritation, or discomfort
 - **Tobramycin: corneal abrasion**
 - **Ciprofloxacin: white crystalline precipitate in patients with corneal defects**
- Antibiotic administration by intracameral injection is available alternatively
 - Injection into anterior chamber of the eye
 - Substantial evidence to support intracameral injection

PHARMACOLOGICAL OPTIONS

ANTI-INFLAMMATORY AGENTS

- **Indications:** reduce inflammation and the risk of developing cystoid macular edema
- Usually initiated the day before or the day after surgery and continued for a few weeks postoperatively (longer if evidence of cystoid macular edema)
- **NSAIDs:** bromfenac, diclofenac, ketorolac, nepafenac
 - NSAIDs alone or in combination with topical corticosteroids have been proven to reduce the incidence of early postoperative cystoid macular edema
- **Corticosteroids:** dexamethasone, difluprednate, fluorometholone, loteprednol, prednisolone
- **Side effects:**
 - **Corticosteroids:** elevated intraocular pressure (IOP), can mask signs of infection
 - **NSAIDs:** mild to moderate burning, possible ulceration with prolonged use, anterior chamber inflammation (bromfenac), eyelid margin crusting, eye pain, punctate keratitis, blurred vision, dry eye, pruritis, headache (nepafenac)

PHARMACOLOGICAL OPTIONS

GLAUCOMA MEDICATIONS

- See [Glaucoma Lecture](#) for list of medications and side effects
- Indication: to reduce intraocular pressure following cataract surgery
- Continue with glaucoma medications if being used prior to cataract surgery

PHARMACOLOGICAL OPTIONS

FYI

DILATORS AND CYCLOPLEGIC AGENTS

- **Indications:** keep iris away from implant during early healing process and improve comfort by decreasing ciliary muscle spasm
 - Less common with modern surgery due to smaller incisions
- Used for a few weeks after surgery
- **Cyclopentolate, phenylephrine, tropicamide**
- **Side effects:** flushing, tachycardia, hypertension, and urinary retention

SAMPLE PHARMACOLOGICAL EYE DROP REGIMEN

NSAIDs

- Decreases pain and inflammation
- Often started 1 day before surgery and continued for a few weeks after

Antibiotics

- Prevent infection
- Can be used pre-, peri-, and post-operatively

Corticosteroids

- Decreases inflammation and when combined with NSAIDs it is more effective at controlling inflammation and preventing early postoperative cystoid macular edema
- Often started after surgery and continued for a few weeks after



Regimens may vary, it is best to consult the practitioner if there is any confusion regarding the eye drop regimen

COUNSELLING TIPS

- **Suspension** eye drops should be **shaken well** before use
 - Remove contact lenses prior to instilling drops & wait at least 15 min after instilling drops to insert contact lens
 - If the applicator of the drop touches a surface, wipe with a clean tissue. Do not rinse tip of bottle with water as this will change the solution's stability and composition
1. Wash hands and pull-down lower eyelid using thumb and index finger to **form a pouch**
 2. Instill only 1 drop and close eye without squeezing or blinking for 30-60 seconds to allow absorption—use **nasolacrimal duct occlusion** by applying gentle pressure with finger across the bridge of the nose to prevent systemic absorption
 3. **Wait 3-5 min between administration of different eyedrops** to prevent flushing away or diluting the first drop—if you are using both eye drops and eye ointment, apply the eyedrop at least 10 minutes before the ointment.
 4. Remove excess fluid gently with a clean tissue—do not rub
 5. Discard bottle **after 28 days** after opening
- Certain diabetic medications (insulin and insulin secretagogues) may be held the day of surgery due to hypoglycemic risk
 - Metformin may be held the day before and day of surgery due to risk of lactic acidosis



Refer to the product monograph for specific instructions on proper use, as information may vary

MONITORING PARAMETERS

Efficacy & Safety Endpoints			
Parameter	Within 24 hours	24 hours – 1 week	>1 week - months
Red flag symptoms (refer within 24 hours)	Severe pain +/- redness	Severe pain, increasing redness, declining vision, retinal detachment (flashes, floaters, visual field defect)	Rapid or significant decrease in vision
Expected symptoms	Decreased vision, mild redness, foreign body sensation	Improving vision, mild redness, improving foreign body sensation, reduced pain	Stabilized vision by 6 weeks, may experience dry eye symptoms long-term

TAKEAWAY

A cataract is an opacity of the lens of the eye that may cause blurred or distorted vision, and in advanced cases, blindness. Cataracts occur frequently with increasing age

Surgery is indicated if symptoms from the cataract interfere with activities of daily living

Topical antibiotics (**tobramycin, fluoroquinolones**) reduce the risk of **endophthalmitis** (bacterial or fungal infection within the eye). Used pre-, peri-, and post-operatively

Anti-inflammatory agents (**NSAIDs, corticosteroids**) reduce the risk of developing **cystoid macular edema**. Used the day before/after and continued for a few weeks post-operatively

Glaucoma agents (e.g. **prostaglandin analogues, beta-blockers, carbonic anhydrase inhibitors, alpha2-adrenergic agonists, cholinergic agonists**) **lower IOP**. Continue if using prior to surgery

Dilators/cycloplegics (**cyclopentolate, phenylephrine, tropicamide**) **keep the iris away from the implant during early healing period**. Used for a few weeks post-operatively

CASE STUDY

KV is a 67-year-old male who has been experiencing distorted and blurry vision, whilst experiencing glare problems over the past 6 months. He is particularly worried as his symptoms have started to affect his ability to drive and he currently works as a truck driver. His past medical history is significant for type 2 diabetes mellitus, hypertension, dyslipidemia, and asthma. His current medications include Synjardy® (empagliflozin 5 mg/metformin 500 mg) PO BID, Twynsta® (telmisartan 40 mg/amlodipine 5 mg) PO daily, rosuvastatin 20 mg PO daily, and Symbicort 200 Turbuhaler® (budesonide 200 mcg/formoterol fumarate 6 mcg) 2 inhalations BID and 1-2 inhalations PRN Q4H. KV reports that he smokes 5 cigarettes a day and drinks 2 glasses of wine daily. KV was diagnosed with cataracts in his right eye and is due to undergo cataract surgery.

CASE STUDY - QUESTIONS

1. Which of the following is NOT a risk factor that KV has for the development of cataracts?
 - a) Type 2 diabetes mellitus
 - b) Age
 - c) Alcohol intake
 - d) Smoking
2. Which of the following counselling points is LEAST appropriate to counsel KV on regarding loteprednol eye drops?
 - a) This medication may be continued for a few weeks post-operatively
 - b) This medication should be started 1 week before cataract surgery
 - c) Adverse effects may include an elevated intraocular pressure and burning at the application site
 - d) Loteprednol is used to reduce the risk of cystoid macular edema (retinal thickening and edema involving the macula)
3. KV will administer the eye drops on his own. Which of the following statements is FALSE with respect to the administration of eye drops?
 - a) The applicator tip should be wiped with a dry tissue before each use
 - b) Applying gentle pressure across the bridge of the nose reduces systemic effects
 - c) Wait approximately 3 to 5 minutes before administering a different eye drop medication
 - d) Eye drops should be discarded 28 days after opening
4. Six days after undergoing cataract surgery, KV is concerned that he is still experiencing mild redness, itchiness, and a foreign body sensation. What is the most appropriate recommendation to make at this stage?
 - a) Refer to the emergency department
 - b) Refer to the ophthalmologist
 - c) Refer to the family physician
 - d) Monitor for dry eyes and symptoms of improvement

REFERENCES

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CHANGE LOG

March 2025

- Formatting changes throughout
- Slide 8: Updated antibiotic use pre-, peri- and post operatively, added intracameral injection alternative, added notes to script
- Slide 9: Added that NSAIDs can reduce the incidence of early postoperative cystoid macular edema
- Slide 13: Updated regimen to reflect changes above, added disclaimer that regimens may vary

August 2025

- Formatting, grammar, spelling changes throughout
- Slide 3: Added legend
- Slide 15: Added monitoring parameters
- Slide 16: Added takeaway
- Slides 17, 18: Added case study and questions
- Slide 19: Updated references

November 2025

- Added 'Post-Operative Care' to title
- No content changes made

February 2026

- Slide 2: Updated NAPRA competencies
- No content changes made